MAULANA AZAD NATIONAL INATITUTE OF TECHNOLOGY, BHOPAL - 462003

B,Tech. (Electrical), 2^{nd}Year

Third Semester

Name of Program	B.Tech (Electrical Engg)	Semester-III	Year-2^{nd}
Name of Course	Mathematics III		
Course Code	MTH-3301		
Core / Elective / Other	Core		

Prerequisite:	
1.	Knowledge of Engineering Mathematics
2.	Basics of statistical concepts such as central tendencies, dispersion etc.

Course Outcomes:	
1.	Apply suitable numerical techniques for real world problem related to Electrical Engineering
2.	Have ability of understanding statistical inferences
3.	Use of Z-transform in signal processing

Description of Contents in brief:	
1.	Numerical Methods: Solution of algebraic and transcendental equations, Solution of linear Simultaneous Equations.
2.	Finite Differences, Interpolation and Extrapolation, Inverse Interpolation.
3.	Numerical Differentiation and Integration.
4.	Numerical solution of Ordinary & Partial Differential Equations.
5.	Statistics: Curve fitting, Correlation and Regression Analysis
6.	Probability Distribution. Sampling and Testing of Hypothesis
7.	Z-transforms, Inverse Z-transforms and properties

List of Text Books:	
1.	Numerical Methods by Dr. B. S. Grewal
2.	Mathematical Statistics-Ray, Sharma and Chaudhary
3.	Advanced Engineering Mathematics by Erwin Kreyszig

List of Reference Books:	
1.	Numerical Analysis by Hildebrand, Mcgraw Hill.
2.	Numerical Analysis by Scarborough, Oxford.
3.	Mathematical Statistics – J E Freund & R E Walpole
4.	Numerical Methods by E .Balaguruswamy, TMH

URLs:	
1.	https://nptel.ac.in/courses/111/107/111107105/
2.	https://nptel.ac.in/courses/111/105/111105041/
3.	https://nptel.ac.in/courses/111/106/111106101/

4.	https://nptel.ac.in/courses/110/107/110107114/
Lecture Plan (about 40-50 Lectures):	
Lecture No.	**Topic**
Solution of Algebraic and Transcendental Equations (10 L)	
1.	Basic Concept and Motivation
2.	Bisection method, RegulaFalsi method, Secant method
3.	General iteration method, sufficient condition for convergence, efficiency of method
4.	Newton-Raphson method, Chebyshev Method
5.	Rate of Convergence and order of convergence of different methods
6.	Newton's method for multiple roots
7.	Problems on previous topics
8.	Direct method; Graffe's Root Squaring Method
9.	Descartes Rule of Signs, Lin Bairstow's Method for finding complex roots
10.	Problems on Graffe's and Lin Bairstow's methods
Solution of Linear and Non-Linear Simultaneous Equations (5 L)	
11.	Direct Methods (Gauss Elimination and Gauss Jordan)
12.	Iterative Method: Jacobi Iterative Method, Gauss-Seidel Method
13.	Factorization Method- LU Decomposition method
14.	Convergence of iterative methods and some problems on previous topics
15.	Newton Raphson method for Non-Linear Simultaneous Equations & its convergence
Finite Differences, Interpolation, Numerical Differentiation and Integration (12 L)	
16.	Forward, Backward and Central difference operators and table
17.	Relation between different operators and some problems
18.	Interpolation with equidistant point: Newton Gregory's Forward interpolation Formula, Backward interpolation formula
19.	Central difference formulae: Gauss Forward, Gauss Backward, Stirling's Formula

20.	Central difference formulae (continued): Bessel's Formula, Laplace-Everette's Formula
21.	Newton Interpolating Polynomial using divided Difference Table
22.	Lagrange Interpolation Polynomial, Lagrange's inverse interpolation
23.	Differentiating tabulated functions, Finding maxima and minima of tabulated functions
24.	Newton-Cotes quadrature formulae: Trapezoidal Rule, Simpson 1/3 Rule
25.	Simpson 3/8 Rule, Weddle Rule
26.	Error analysis
27.	Double Integration
Numerical Solution of ODE and PDE (12 L)	
28.	Picard's method, Taylor's Series (ODE)
29.	Euler and Euler's Modified method (ODE)
30.	Error in these methods and problems (ODE)
31.	Runge- Kutta Method; III & IV order methods, (ODE)
32.	Problems on these methods (ODE)
33.	Milne's Method and Error Analysis (ODE)
34.	Adams Bash-forth Method and Error Analysis (ODE)
35.	Problems on Milne's and Adams Bash-forth methods (ODE)
36.	Classification of second order PDE's, Finite difference approximations to derivatives
37.	Solution of Laplace and Poisson's Equations (case of elliptic equations)
38.	Solution of Heat equation (case of parabolic equation)
39.	Solution of Wave equation (case of hyperbolic equation)
Statistics (10 L)	
40.	Curve Fitting by Method of Least Squares
41.	Correlation: concept, Type of correlation, Coefficient of correlation, Rank correlation
42.	Regression: concept, Line of regression of X on Y, Line of regression of Y on X,
43.	Relation between correlation and regression, Standard Error of Estimate

44.	Binomial and Poisson Distribution (Discrete)
45.	Normal Distribution (Continuous)
46.	Introduction to Sampling, Sampling Distribution and important terminology
47.	Introduction to Testing of Hypothesis related terminology
48.	Z-test
49.	t-test, Chi-square test
50.	Z-transforms, Inverse Z-transforms and properties.

Name of Program	**B. Tech (Electrical)**	**Semester- III**	**Year- 2nd**
Name of Course	**Fundamentals of Design Methods**		
Course Code	**ME 3302**		
Core / Elective / Other	**Core**		

Prerequisite if any:

1.	Strength of Material
2.	Engineering Mechanics

Course Outcomes:

1	An ability to observe the behavior of forces on a component, formulate the problem and draw the design specifications.
2	An ability to understand and formulate the behavior of component subjected to loads and identify the failure criteria.
3	An ability to analyze the stresses and strains induced in a machine element.
4	An ability to design a machine component using theories of failure.
5	An ability to design pins joints, cotters, couplings and joints including riveted, bolted and welded joints.
6	An ability to find the feasible solution *(with creative aptitude)*for a given problem with the help of design phases.
7	An ability to use the standard data for designing the component.

Description of Contents in brief:

1.	**Introduction-** Engineering Design and classification. Basic design procedure, requirement of machine element, traditional design methods. Material selection on the

	basis of material index. Concept of methods- preliminary design, conceptual design, detailed design. Concurrent engineering, reverse engineering, and creative design. Factor of safety. Concept of Machine Design: Types of loads. Simple Stress and Strain.
2.	**Fasteners-** Types of fasteners. Types of rivet joints, rivet heads, terminology, caulking and fullering, riveted joints (Lap and Butt). Design of welded joints: Types of welded joints Strength of parallel and fillet weld, strength of a welded joint.
3.	**Threaded joint-** Selection of standard threaded joint (Brackets) eccentric and offset loading.
4.	**Pin type joints-** Standard pin type joints, Design Procedure on failure criteria and empirical relation based- Knuckle and cotter.
5.	**Shaft-** Design of shaft, Design of shaft couplings rigid, flange and protective type flange. Design for power transmission and screws Effects of stress concentration in design.
6	**Springs-** Design of springs: Helical springs, closed and open coiled tension, compression springs and their ends, design of leaf springs.
7	**Design of power screws-** designing for various types of screw jacks, lead screw of lathe machine and screw press. Computer aided design of at least one joint.

List of Text Books:

1.	Mechanical Engineering Design	J.E. Shigley and Charles R. Mischke, TMH
2.	Engineering Design	G.E. Dieter
3.	Machine Design	Kulkarni, TMH
4.	Design of Machine Elements	V B Bhandari TMH
5.	Machine Design	R. L. Norton Tata McGraw Hill, 2005.

List of Reference Books:

1.	Mechanical Engineering Design: Principles and concepts,	Siraj Ahmed (2014): PHI Learning Pvt. Ltd, ISBN-978-81-203-4931-5, New Delhi
2.	The Mechanical Design Process.	David G. Ullman: ISBN 978–0–07–297574–1 McGraw- Hill, Inc. N.Y.

URLs:

1.	https://nptel.ac.in/courses/112/105/112105124/

Lecture Plan (about 40-50 Lectures):

***Lecture No.**	**Topic Remarks**
1.	Introduction of subject, Simple stresses.
2.	Concept of critical section of a part.
3.	Brief discussion on combined stresses.
4.	Material selection approaches and methods.
5.	Material Selection method (Digital logic method).
6.	Concept of preliminary design, conceptual design and detailed design.
7.	Concept of concurrent engineering, reverse engineering, and creative design. Creative techniques.
8.	Simple Stress and Strain.

9.	Applied problems.
10.	Introduction of fasteners, Types of Fasteners.
11.	Design Procedure of knuckle joint (failure criteria and empirical relation).
12.	Knuckle joint applied problems (failure criteria).
13.	Knuckle joint applied problems (empirical relation).
14.	Design Procedure of cotter joint (failure criteria and empirical relation).
15.	Tutorials
16.	Cotter joint applied problems (failure criteria).
17.	Cotter joint applied problems (empirical relation).
18.	Design Procedure of riveted joint.
19.	Applied problems on riveted joint.
20.	Design Procedure of welded joints.
21.	Applied problems on welded joints.
22.	Design Procedure of threaded joint (In plane).
23.	Applied problems on threaded joint (In plane).
24.	Design Procedure of threaded joint (offset).
25.	Applied problems on threaded joint (offset).
26.	Tutorials
27.	Design of shaft based on combined stresses.
28.	Applied problems on shaft.
29.	Design of rigid couplings.
30.	Applied problems on rigid couplings.
31.	Design of flange couplings.
32.	Applied problems on flange couplings (non-protective).
33.	Tutorials
34.	Applied problems on flange couplings (non-protective).
35.	Tutorials
36.	Power screw and their types. Design of power screw.
37.	Applied problems on Power screws.
38.	Design of power screw (stresses on power screw).
39.	Applied problems on Power screws.
40.	Tutorials
41.	Design of springs (Stress and deflection equation based).
42.	Materials and material selection for springs.
43.	Applied problems on static and impact load.
44.	Applied problems on static and impact load.
45.	Design of spring against fluctuating load.
46.	Applied problems on fluctuating load
47.	Tutorials
48.	Design of leaf spring.
49.	Applied problems on leaf spring.
50.	Applied problems on leaf spring.

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Name of Program: B.Tech (Electrical)	Semester : III	Year: 2nd
Name of Course:	Electrical Engg. Materials & EM Fields	
Course Code:	EE3303	
Core/Elective/other:	Core	
Prerequisite: Fundamentals of Physics (Electricity), Vectors, Vector Calculus, Differential Equations		
Course Outcome: After studying this subject, students will be able to		
01	*apply concepts of Electric fields in Electrical Devices & power Networks*	
02	*apply concepts of Magnetic fields in Electrical Devices& power Networks*	
03	*select Electrical Materials for specific application in Electrical Engineering*	
Description of Contents in brief		
1	Review of vectors and 3D coordinate systems, Electrostatic Fields -Coulomb's law, electric field intensity, Gauss's Law, physical concepts of divergence, gradient and curl.	
2	Electric potentials, boundary relations for electric fields, capacitance, continuity equation, Poisson's and Laplace's equations	
3	Magnetic Fields - Biot–Savart's Law, Stoke's theorem, Ampere's circuital law, magnetic boundary conditions, energy storage in magnetic fields, Scalar and vector magnetic potential, Forces & torques on current loops	
4	Maxwell's equations, EM Wave, Propagation, Poynting theorem, instantaneous and average power, Polarization of waves, reflection and refraction of waves,	
5	Insulating & Dielectric Materials, Conducting Materials, Magnetic Materials, superconductivity.	
List of Text Books		
01	Sadiku, Elements of Electromagnetics, 6 edition, Oxford University Press	
02	William &Haytt, Engineering Electromagnetic, 8th edition, McGraw Hill Education	
03	Kraus, Electromagnetics With Applications, Ninth Reprint, Indian Edition, McGraw Hill	
04	Alagappan, Electrical Engineering Materials, Tata McGraw-Hill Publisher	
05	C.S. Indulkar, Electrical Engineering Material, 4th Edn., S Chand & Company.	
List of Reference Books		
01	Jorden &Balmain, Electromagnetic Waves & Radiating Systems, 2nd edition, PHI publication	
02	I.E. Irodov, Basics Laws of Electromagnetics, CBS Publishers India	
03	Dekker, Electrical Engineering Materials, PHI Publications, Eleventh Indian Reprint.	
URLs	http://nptel.ac.in (for "Electromagnetic fields")	
Lecture Plan (about 40-50 lectures)		
Lecture No	Topic	
1	Introduction to 3D coordinate systems	
2	Review of vectors	
3	Transformation of vectors	
4	Various charge distribution, Coulomb's Law, Numerical Problems	

5	Determination of Static Electric fields
6	Numerical Problems
7	Application of Gauss Law, Maxwell's first Equation.
8	Numerical problems
9	Concepts of divergence
10	Numerical problems
11	Electric potential
12	Numerical problems
13	Concepts of gradient, Numerical problems
14	Concepts of Curl, Numerical problems
15	Boundary relation's of Electric fields
16	Numerical Problems
17	Capacitance, Laplace's & Poisson's Equations
18	Numerical problems on Laplace's & Poisson's Equations.
19	Equation of Continuity, point form of Ohm's law
20	Electric Dipole, Numerical problems
21	Method of Images, Numerical problems
22	Properties of Magnetic fields, Maxwell's second equation, Biot-Savart Law
23	Determination of magnetic field intensity
24	Numerical problems on magnetic field intensity
25	Numerical problems on magnetic field intensity
26	Calculation of inductance of various elements
27	Amper's Circuital law and applications
28	Maxwell's third equation and its modification
29	Lorentz force and numerical problems
30	Torque on current carrying loop, Numerical problems
31	Concept of Scalar and vector magnetic potential
32	Energy stored in magnetic fields, Numerical problems
33	Boundary relations of magnetic fields
34	Numerical problems on Boundary relations
35	Faraday's Law and Maxwell's fourth equation

36	Maxwell's equations in various forms
37	Poynting Theorem
38	Instantaneous & Average Power
39	Numerical problems
40	Uniform Plane waves
41	Wave equations, solution, and behavior in various mediums
42	reflection and refraction of waves
43	Polarization of waves
44	Properties of insulating and dielectric materials
45	Application of insulating and dielectric materials
46	Properties of conducting materials
47	Application of conducting materials
48	Classification and properties of magnetic materials
49	Application of Magnetic materials
50	Superconductivity and its application

Name of the Program:	**B.Tech.**	Semester	**Third**	Year	**2nd**
Name of the Course:	EMEC-I				
Course Code:	EE-3304				
Core/Elective/Other:	Core				
Pre-requisites:	Fundamentals of Basic Electrical Engineering				
Course Outcomes:					
1	Understanding of concept of electromechanical energy conversion				
2	Develop the understanding of polyphaser circuits				
3	Detailed understanding of polyphaser transformers				
4	Understanding of DC generators and motors				
Description of Content in Brief:					
1	**Electromechanical energy conversion:** Forces and torque in magnetic systems, energy balance, energy and force in a singly excited and multi excited magnetic field systems, determination of magnetic force, coenergy				
2	**Polyphase circuits:** Three-phase systems with balanced & unbalanced load, measurement of power and power factor in three phase circuits				

3	**DC Generators:** Principle of operation, construction, lap and wave windings, emf equation, types of DC generators, voltage buildup, critical field resistance and critical speed, Armature reaction and commutation, methods of improving commutation, Characteristics, parallel operation
4	**DC Motors:** Principle of operation, back EMF, torque equation, types of DC motors, characteristics, methods of speed control, types of starters, Losses and efficiency
5	**Polyphase Transformers:** Review of single phase transformer,Three phase transformer: Principle of operation, vector groups, open delta connection, Scott connection, Auto transformers, harmonic reduction in phase voltages.
6	**Testing of transformers and DC machines:** Testing of DC machines: Swinburne's test, Hopkinson's test, OC/SC and Sumpner's test on transformer
List of Text Books:	
1	P.S.Bhimbra, Electrical Machine, Khanna, 1 January 2011
2	Nagrath& Kothari, Electrical Machines, 4 edition, McGraw Hill Education, 7 July 2010
List of Reference Books	
1	M. G. Say, Performance & design of A.C. Machines, 3rd edition , CBS, 1 December 2005
2	Fitzerald Kingsley Otmans, Electrical Machines, 7 edition , McGraw-Hill Education, 1 March 2013
3	Charles. I. Hubert, Electric Machine, 2 edition, Prentice Hall, 16 October 2001
4	J.R. Cogdell, Foundation of Electric Power, Pck edition , Addison Wesley Longman, May 1, 2003
5	M. G. Say, Performance & design of A.C. Machines, 3rd edition , CBS, 1 December 2005
URLs:	
1	https://nptel.ac.in/courses/108/105/108105017/
Lecture Plan	
1.	Introduction to polyphase system, Generation of 3-phase voltages
2.	Star and Delta connected network, Relation between phase quantity and line quantity
3.	Three phase power, Numericals on star and delta connections
4.	Numericals
5.	Measurement of 3-phase power: One, two and three wattmeter method
6.	Measurement of power factor using two wattmeter method
7.	Analysis of unbalance load
8.	Numericals-1
9.	Numericals-2
10.	Elements of Electro-Mechanical Energy Conversion-1
11.	Fundamentals of Energy balance, Magnetic Field System: Energy and Co-Energy.
12.	Singly and doubly excited system-1

13.	Numericals-1
14.	Constructional features of D.C machine.
15.	Classification of D.C. Machine. Type of windings.
16.	E.M.F equation of D.C. Generator.
17.	Losses in D.C Generator, Power stages and efficiency.
18.	Armature reaction and its remedies-1
19.	Armature reaction and its remedies-2
20.	Numericals
21.	Commutation and methods to improve it-1
22.	Commutation and methods to improve it-2
23.	Generator Characteristics-1
24.	Generator Characteristics-2
25.	Numericals-1
26.	Numericals-2
27.	Motor principle, significance of back E.M.F., Torque Equation.
28.	Motor characteristics-1
29.	Motor characteristics-2
30.	Losses and efficiency.
31.	Numericals-1
32.	Numericals-2
33.	Starting of D.C. machine-1
34.	Speed control of D.C. machine-1
35.	Speed control of D.C. machine-2
36.	Testing of D.C machine-1
37.	Testing of D.C machine-2
38.	Numericals
39.	Review of 1-phase transformer.
40.	Open Delta connection
41.	Scott Connection
42.	Connections of 3-phase transformer-1
43.	Connections of 3-phase transformer-2
44.	Parallel operation of 1-phase transformer-1
45.	Parallel operation of 1-phase transformer-2

46.	Auto transformer-1
47.	Auto transformer-1
48.	Numericals
49.	OC/SC and Sumpner's test
50.	Numericals

Name of the program	B.Tech. Electrical	Semester- III	**Year -2nd**
Name of Course:	NETWORK Analysis		
Course Number:	**EE 3305**		
Core/Elective/other	**Core course**		
Pre-requisites:			
1	Differential equation		
2	Laplace Transform		
3	Basic electrical		
Course Outcomes			
1.	To analysis networks		
2.	To get the concepts of circuit and synthesis		
3.	To apply basic theorems to electrical circuits		
Contents			
1.	Circuit Concept, Network Topology, Laplace transform, Waveform synthesis, Fourier series		
2.	Steady State Analysis, Transient Analysis, Initial conditions		
3.	Network theorems		
4.	Driving point & transfer impedances, Network function-Poles &zeros, two port networks		
5.	Conditions for realizing an immittance function of passive elements, Foster form &Cauer form of RC, RL & LC networks		
6.	Coupled circuits, Network synthesis		
Text Books:			
1.	Van Valkenberg, Network Analysis, Prentice-Hall,		
2.	S.Ghosh, Network theory: analysis and synthesis, PHI,		
3.	Pankaj Swarnkar, Network Analysis and Synthesis, SatyaPrakashan		

4.	B. S. Manke, Network Analysis, Khanna Publisher
Reference Books :	
1.	Schaum's Outline Series, Circuit theory,
2.	C.L.Wadhwa, Network Analysis & Synthesis, New Age International,
URL	
1.	https://nptel.ac.in/courses/108105159/
2.	https://nptel.ac.in/courses/108102042/
3.	https://www.classcentral.com/course/swayam-network-analysis-17705
	Lecture Plan
Lecture No.	**Topic To Be Covered**
4.	Introduction Voltage/Current Relationship For Individual Element
5.	Circuit Concepts RLC Parameters
6.	Independent and Dependent Sources, Source Transformation
7.	Tutorials
8.	Transient And Steady State Analysis
9.	Step Function Response To RL, RC And RLC Network And Concept Of Time Constant.
10.	Initial Conditions In Networks
11.	Problems on Initial Conditions
12.	Problems on Initial Conditions
13.	Laplace Transforms Basic Theorems of Transient Analysis
14.	Tutorials on Laplace Transforms
15.	Waveform Synthesis
16.	Transform of Pulse And Periodic Functions.
17.	Tutorials on waveform synthesis
18.	Network Topology : Graph Tree, Co-Tree Incidence Matrix
19.	Tieset, Cut-Set, Loop Impedance Matrix
20.	Tutorials on Graph Theory
21.	Formulation Equilibrium Equation Based On Loop/Node Basis
22.	Poles &Zeros of Transfer Function
23.	One and Two Port Networks
24.	Various Two Port Network Parameters
25.	Interconnections of Two Port Network
26.	Tutorials on Two Port Networks
27.	Tutorials on Two Port Networks
28.	Network Theorems
29.	Superposition, Millman Theorem
30.	Tutorials on Theorems
31.	Thevenin's, Norton's Theorem
32.	Maximum Power Transfer Theorems
33.	Tutorials on Theorems
34.	Frequency Domain Analysis - Fourier Series,

35.	Expansion of Fourier Series, Phase And Amplitude Spectrum,
36.	Sinusoidal Steady State Analysis, Phasors and Phasor Diagram,
37.	Exponential Form, Fourier Integral & Transforms
38.	Relationship Between Laplace & Fourier Transforms.
39.	Tutorials on Fourier Series
40.	Mutual Inductance
41.	Coupled Circuits.
42.	Tutorials on Coupled Circuit
43.	Tutorials on Coupled Circuit
44.	Introduction to Network Synthesis. Positive Real Function
45.	Driving Point & Transfer Impedance,
46.	Foster Form of RC, RL & LC Networks
47.	Cauer Form of RC, RL & LC Networks
48.	Tutorials on Foster and Cauer Forms
49.	Tutorials on Foster and Cauer Forms

Fourth Semester

Name of Program	Electrical Engineering	Semester IV	Year: 2^{nd}
Name of Course	MATHEMATICS-4		
Course Code	MTH-3401		
Core/Elective/Other	CORE		
Prerequisite:			
1.	Basic concepts of algebra and arithmetic.		
2.	Basic concepts of maximization and minimization of functions.		
3.	Curiosity to formulate and solve real world optimization problem.		
Course Outcomes:			
1.	Ability to understand the importance of operations research in decision making and use them effectively in different situations.		
2.	Ability of modeling, solving and analyzing optimization problems by using Operations Research.		
3.	Ability to understand the characteristics of different types of decision-making environments and the appropriate decision making approaches and tools to be used .		
Description of Contents in Brief:			
1.	Linear Programming, Graphical method, Simplex algorithm, Duality, Transportation problem, Assignment problem.		
2.	Duality theory, Dual Simplex method, Revised simplex method, Sensitivity analysis. Integer programming- Cutting plane algorithm, Branch and bound technique, Travelling salesman problem..		
3.	Nonlinear programming, Kuhn- Tucker conditions, quadratic programming, Wolfe's algorithm. Dynamic programming- Deterministic and stochastic example.		
4.	Queuing Theory, Queuing Problem and System, Transient and Steady State Distributions in Queuing System, Poisson Process, Exponential Process, Classification of Queuing Model.		
5.	Games Theory, Minimax-Maxmin Principle for Mixed Strategy Games, Graphical Method, Solution of (MxN) Job Sequencing, PERT – CPM.		
List of Text Books:			
1.	Operations Research, S. D. Sharma, KedarnathRamnath& Sons. 14/e		
2.	Optimal Trading Strategies by RobetKissell, Morton Glantz		
3.	Numerical Optimization Techniques with Applications by Suresh Chandra		

List of Reference Books:	
1.	Operations Research, Gupta and Hira, S.Chand Publication
2.	Introduction to Optimization Operations Research, J.C. Pant, Jain Brothers
3.	Introduction to Operations Research- Hillier and Lieberman,McGraw-Hill
4.	Operations Research An Introduction- TahaHamdy,Prentice-Hall
URLs:	
1.	http://www.pondiuni.edu.in/storage/dde/downloads/mbaii_qt.pdf
2.	https://www.thelearningpoint.net/home/mathematics/an-introduction-to-operations-research
3.	nptel

Lecture Plan

Lecture No.	**Topic**
1	Linear Programming (LPP) – Introduction
2	Formulation of LP Problem
3	Graphical method of LPP
4	Simplex algorithm of LPP
5	Two phase method
6	Big-M Method
7	Degeneracy in LPP
8	Duality in LPP
9	Transportation problem (TP): Introduction, Mathematical Formulation, Matrix Representation of TP
10	Initial Basic Feasible Solution of Transportation problem : North West Corner Rule, Row Minima Method
11	Column Minima Method, Matrix Minima Method
12	Vogel’s Approximation Method
13	Optimality test for TP – UV Method
14	Unbalanced Problem of TP
15	Assignment problem: : Introduction, Mathematical Formulation, Matrix Representation of Assignment Problem
16	Hungarian Method for Assignment problem
17	Unbalanced Problem
18	Maximization Problem
19	Duality theory
20	Dual Simplex method
21	Revised simplex method
22	Sensitivity analysis
23	Integer programming (IPP)- Cutting plane algorithm
24	Branch and bound technique for IPP
25	Travelling salesman problem
26	Nonlinear programming: Introduction
27	Kuhn- Tucker conditions
28	Illustrative examples on Kuhn- Tucker conditions
29	Quadratic programming
30	Wolfe's algorithm
31	Dynamic programming : Introduction, Bellman’s Principle of Optimality
32	Minimum Path Problem in Dynamic Programming
33	Single additive constraint, multiplicatively separable return

34	Single additive constraint, additively separable return
35	Deterministic and Stochastic examples
36	Queuing Theory: Introduction, Terminologies, List of Symbols
37	Queuing Problem and System
38	Transient and Steady State Distributions in Queuing System
39	Poisson Process
40	Exponential Process
41	Classification of Queuing Model
42	Games Theory: Introduction, Characteristics, Basic Definitions
43	Minimax-Maxmin Criterion and Optimality Strategy
44	Principle for Mixed Strategy Games
45	Graphical Method
46	Solution of (Mxn) Job Sequencing
47	Practice Problems on Job Sequencing
48	Project Management : Introduction, Basic Definitions, Network Diagram
49	CPM
50	PERT

Name of the Program:	**B.Tech. (Electrical)**	Semester	**IV**	Year	**2nd**
Name of the Course:	Fundamentals of Entrepreneurship				
Course Code:	HUM-3402				
Core/Elective/Other:	Core				
Pre-requisites:	1. Fundamental Understanding of Personnel Administration is an integral part for Engineers before entry into organization and to setup any startup business. 2. Organizational theories and Motivation for Entrepreneurship enthusiastic for systematic approach to handle the materialistic and non-materialistic motivation				

Course Outcomes:	
1	Course will inculcate administrative skills for Engineers to handle the all the matters related to personnel and their dimensions.
2	Engineering students will be in position to take appropriate decision to understand the industrial Laws and application of administrative principles for smooth functioning of Industry and a fresh idea to setup new business.
3	Boost up the engineers to develop Business ideas to become successful entrepreneurs to setup new operational business units to address the need of society.
Description of Content in Brief:	
1	**Introduction:** Meaning, nature and scope of Personnel Administration in India, functions and significance of Personnel Administration, Recruitment, Training, Promotion and Disciplinary Action, Classification of Services, Generalists and Specialists, Development of Public Services in India, Bureaucracy and Modern Democratic System, Performance Appraisal.

2	**Organizational Theories:** Organizational Development Theories, Fredric Winslow Taylor, Marry Parker Follett, Elton Mayo, Max Weber, Henry Fayol, Power, Accountability, Responsibility, Control, Transparency and Conflict Resolutions
3	**Entrepreneurship:** Meaning and importance of Entrepreneurship, Evolution of Entrepreneurship, Factors influencing Entrepreneurship: Social factors, psychological factors, economical factors and environmental factors. Characteristics of an Entrepreneur, Types of Entrepreneur: type of business, use of technology, motivation. New Generation of entrepreneurship: social entrepreneurship, tourism entrepreneurship, women entrepreneurship. Barriers to Entrepreneurship.
4	**Entrepreneurial motivation:** Relevance of Motivation, Maslow's Theory, Herzberg's Theory, Douglas McGregor, Ethics, Corruption and Anti-Corruption Machinery in country
5	**Industrial Laws:** Factory Act 1948, Provident Fund Act 1952, Compensation Act 1919, Special Economic Zone (SEZ), National Small Industries, Quality Standard with Special reference to (ISO), Small Industries Development Bank of India (ISDBI), New Pension Scheme 2004 Act.
List of Text Books:	
1	D. Ravindra Prasad, V.S. Prasad, P. Satyanarayana, Y. Pardhasaradhi. Administrative Thinkers. Sterling Publishers Private Limited Second Revised Enlarged Edition, New Delhi. 2010.
2	Haidi, Patricia & Brush, Teaching Entrepreneurship, Edward Elgar, United States of America, 2014.
3	Norma M. Riccucci, Public Personnel Management, Rutledge, United States of America, 2017.
4.	Padhi, Labour& Industrial Laws, PHI Learning Private Limited, Eastern Economy Edition. New Delhi,2017
List of Reference Books	
1	Monappa&Saiyadain, Personnel Management, McGraw Hills, January-2001.
2	Howard & Donald, Entrepreneurship: Theory, Process &Practice, engage Learning, Australia, 2010.
3	Jared, Donald & John, Public Personnel Management, Rutledge, United Kingdom,2018
4	Sinha &Sekhar, Industrial Relations, Trade Unions and Labor Legislations, Pearson Publication, New Delhi.
URLs:	
1	https://www.my-mooc.com/en/categorie/entrepreneurship
2	https://nptel.ac.in/courses/110/106/110106141/
3	https://nptel.ac.in/courses/122/105/122105020/
Lecture Plan	
1.	Historical Background of Organization and its Personnel with its nature and scope.
2.	Meaning definitions, relevance of personnel administration in organizations
3.	Why to understand personnel administration first than entrepreneurship development for engineers
4.	Types of organizations and personnel administration
5.	Laws of recruitment, types, legal procedures in the organization to promote and demote the personnel, Performance Appraisal

6.	Development of personnel administration in India, bureaucracy and its types in India, Contribution of Max Weber
7.	Promotion, types of promotion, disciplinary action and classification of services, Administrative tribunals
8.	Generalists Vs. Specialists, their arguments and need of the hierarchy at the top of the organization.
9.	Modern Democratic system and organizations in the world
10.	Organizational Theories and its development; Contributions of classical thinkers in the professionalism.
11.	Contributions of Frederick Winslow Taylor in the development of Scientific management approach, functional foremanship and development of scientific tools and techniques for more efficiency.
12.	Top level management and contribution of French engineer in the development of basic professional principles for high rise in production.
13.	Hawthorne Experiment and emergence of new Human Relation approach by the Elton Mayo
14.	Marry Parker Follet and orders to follow in different situations and understand the behavior of employees.
15.	Accountability, Responsibility, Power, Control, Authority and its Impact on people.
16.	Transparency and conflicts in the organizations
17.	Interdependence between the Entrepreneurship and Personnel administration
18.	Basic understanding of entrepreneurship ,its relevance and challenges
19.	Factors influencing the entrepreneurship
20.	Social factors, psychological factors, economical factors and environmental factors
21.	Characteristics of an Entrepreneur,
22.	Types of Entrepreneur: type of business, use of technology, motivation
23.	New Generation of entrepreneurship: social entrepreneurship, tourism entrepreneurship, women entrepreneurship.
24.	Barriers to Entrepreneurship
25.	Entrepreneurial motivation: Relevance of Motivation to start a new business
26.	Maslow’s Need Hierarchy Theory, Herzberg’s Theory X and Y.
27.	Douglas McGregor, Ethics,
28.	Corruption and Anti-Corruption Machinery in country
29.	Factory Act 1948, Provident Fund Act ,1952
30.	Compensation Act, 1919 and its implementation in the industries
31.	Provident Fund Act, 1952 its provisions and benefits.
32.	Rural Entrepreneurship in India
33.	Quality Standard with Special reference to (ISO)

34.	New Pension Scheme 2004,Act for all the central and states government employees
35.	Industrial Development Bank of India (IDBI)
36.	Case studies on Entrepreneurship in the world and India
37.	National Institute of Entrepreneurship in India
38.	Banking system in India and Entrepreneurship
39.	Central Government Initiatives to start a business
40.	Problems and Prospects of entrepreneurship in India
41.	Special Economic Zone (SEZ)
42.	Industrial Dispute Act,1947
43.	Trade Union Act, 1926
44.	Equal Remuneration Act, 1976
45.	Inter-state Migrant Workmen Act,1979
46.	Child Labor (Prohibition & Regulations) Act,1986
47.	Sexual Harassment at the Workplace (Prevention, Prohibition & Redresses) Act, 2013
48.	Contract Labor (Regulation & Abolition) Act,1970
49.	Apprentices Act, 1961
50.	Case Studies

Name of Program: B.Tech (Electrical)	Semester : IV	Year: 2nd
Name of Course:	Generation of Electrical Power	
Course Code:	EE3403	
Core/Elective/other:	Core	
Prerequisite: Fundamentals of Electrical Machines, Prime movers, Engineering Mathematics		
Course Outcome: After studying this subject, students will be able to		
01	Understand structure of modern power systems	
02	explain the basic principle of Generation of Electrical power from various sources	
03	analyze economics of power plants	
Description of Contents in brief		
1	Introduction to electrical power generation and classification of conventional & nonconventional power generation, principle of operation of solar, wind and tidal power, cogeneration, load curves, performance factors	
2	tariffs, basic structures of electricity markets, power factor improvement by static and synchronous capacitors, concept of reactive power compensation, concept of demand side management.	

3	thermal Power Stations: Choice of coal fired station site, arrangement of plant and principal auxiliaries, coal handling plant,
4	Nuclear Power Stations, nuclear physics, site selection of nuclear stations, classification of reactors, working principle of reactors.
5	Hydro-Electric Stations, site selection of hydro stations, hydrology, mass curve, flow duration curve, water storage, classification, main components, pumped storage plants.
6	Economic Aspects of Power Plant Operation Economic Scheduling of Power Stations, Hydro-thermal coordination.
List of Text Books	
01	B.R. Gupta, Generation of Electrical Energy, S Chand publisher
02	M.V.Deshpande, Elements of power station design, Prentice Hall India Learning Private Limited.
03	G.D. Rai, Non-conventional Energy Sources, Khanna Publishers
List of Reference Books	
01	G.R. Nagpal, Power Plant Engineering, 15th Edition, Khanna Publishers
02	L. K. Kirchmayer, Economic operation of power systems, Wiley Eastern Ltd.
URLs	
1.	http://nptel.ac.in (for “Electrical Engineering Energy Resources & Technologies”)
Lecture Plan (about 40-50 lectures)	
Lecture No	Topic
1	Basic architecture of modern power systems
2	Generation of power from various sources in India
3	Significance of factors like connected load, MD, LF, GDF, UF etc.
4	Numerical problems
5	Power factor improvement, condition for economic power factor correction
6	Power factor by static and synchronous capacitors,
7	Numerical problems
8	Active and reactive power control
9	Capability curves of generators
10	Basic theory of reactive power compensation: Load & System compensation
11	Economic Aspects of Power Plant Operation-Fixed charges, interest and depreciation charges
12	methods of depreciation, straight line and sinking fund methods
13	Numerical problems
14	Different types of tariffs & Concepts of real time pricing
15	Numerical problems
16	Basic structure of electricity markets
17	Concepts of Demand side management
18	Choice of coal fired station site, arrangement of plant and principal auxiliaries.
19	Coal handling plant, Ash handling plant.
20	Heat recovery equipments

21	Steam flow diagram
22	Speed governor system power plant instrumentation.
23	Nuclear Power Stations-Nuclear Physics, Atomic energy fuels, site selection
24	Moderator materials, fissile and fertile materials
25	Fission & Fusion reactions
26	Types of reactors, principal parts of nuclear power plant
27	Operation and control of reactors
28	General discussion on various types of conventional & non-conventional power generation
29	Hydro-Electric Stations-Choice of site
30	Arrangement of hydroelectric installations
31	Hydrology, Mass curve, flow duration curve, water storage
32	Classification of hydro electric plants
33	Pumped storage plants, operating cost of hydroelectric station
34	Renewable power and distributed generation
35	Wind and solar power generation
36	Tidal power generation, mini-micro hydro power stations
37	Cogeneration
38	Thermal plant input/output curves, heat rate, incremental cost
39	Economic Scheduling of Power Stations
40	Economic operation of power systems
41	Criteria for loading of power plants without transmission loss
42	Criteria for loading of power plants with transmission loss
43	Coordination equations
44	Iterative methods to solve coordination equations,
45	Numerical problems
46	Load dispatching in power system cogeneration, Base & peak load operation
47	Hydro thermal coordination
48	Methods for computing the generation scheduling in combined hydro thermal system

Name of Program: B. Tech	Semester : IV	Year: 2^{nd}
Name of Course:	EMEC-II	
Course Code:	EE3404	
Core/Elective/other:	Core	
Prerequisite: Concept of rotating electrical machines, Transformers and DC Machines		
Course Outcome:		
1	Understand the constructional details and principle of operation of AC Induction and Synchronous Machines.	

2	Analyze the performance of the AC Induction and Synchronous Machines using the phasor diagrams and equivalent circuits.
3	Select appropriate AC machine for any application and appraise its significance
Description of Contents in brief	
1	**Poly-phase induction motors-** Construction, Principle, Operation and applications, No-load and blocked rotor test, equivalent circuit, circle diagram, Starting & speed control of Induction Machines, . Induction generators – types, principle of operation, equivalent circuit and applications.
2	**Alternators-** Construction, Principle and Types, armature windings, E.M.F. equation - armature reaction, load characteristics, OC & SC tests, voltage regulation – two-reaction theory – parallel operation, parallel operation, operation on infinite bus. Two reaction theory, power expressions for cylindrical and salient pole machines.
3	**Synchronous motors-** Principle of operation, starting methods, phasor diagram, torque angle characteristics V-curves, power factor control of synchronous motor, Applications.
4	**Single Phase Induction Motor**-Double revolving field theory, equivalent circuit, no load & block rotor tests, starting methods.
5	IS codes for testing of AC machines and their applications in electrical systems.
List of Text Books	
1	Nagrath, I.J. and Kothari, D.P., 'Electrical Machines', Tata McGraw Hill Education Private Limited Publishing Company Ltd., 4th Edition, 2010.
2	Dr. P.S. Bhimbra, 'Electrical Machinery', Khanna Publications, 7th Edition, 2007.
3	M. G. Say, 'Performance and Design of Alternating Current Machines', CBS Publishers & Distributors Pvt. Ltd., New Delhi, 3rd Edition, 2002.
List of Reference Books	
1	Arthur Eugene Fitzgerald and Charles Kingsley, 'Electric Machinery', Tata McGraw Hill Education Publications, 6th Edition, 2002.
2	Parkar Smith, N.N., 'Problems in Electrical Engineering', CBS Publishers and Distributers, 9th Edition, 1984.
3	Miller, T.J.E., 'Brushless Permanent Magnet and Reluctance Motor Drives', Clarendon Press- Oxford, 1989.
URLs	
1.	https://nptel.ac.in/courses/108/106/108106072/

Lecture Plan

Lecture No	Topic
L-1	Introduction of Poly Phase induction machine
L-2	Classification and construction of induction machine
L-3	Operating principle of three phase induction motor
L-4	Flux and mmfPhasors of three phase induction motor, rotating magnetic field
L-5	Rotor emf current and power
L-6	Power flow diagram of 3-phase induction motor

L-7	Losses and efficiency of 3-phase induction motor
L-8	Induction motor phasor diagram
L-9	Equivalent circuit diagram
L-10	Numerical problems related to power flow in induction motor
L-11	Analysis of equivalent circuit diagram
L-12	Numerical Problems associated to equivalent ckt diagram
L-13	Torque slip characteristics of induction motor
L-14	Effect of rotor resistance on starting and maximum torque and operating slip
L-15	Numerical problem related to torque slip of induction motor
L-16	Power slip characteristics of induction motor
L-17	Effect of rotor resistance on maximum power and operating slip
L-18	Numerical problem Power slip characteristics of induction motor
L-19	Analysis of operating performance of induction motor
L-20	No-load and block rotor test on induction motor
L-21	Numerical problem No-load and block rotor test on induction motor
L-22	Analysis of induction motor by Circle diagram method
L-23	Numerical problem on Analysis of induction motor by Circle diagram method
L-24	Power factor control of induction motor
L-25	Different method of Speed control of induction motor
L-26	Speed control by pole changing method
L-27	Speed control by stator voltage control
L-28	Speed control by supply voltage and supply frequency
L-29	Speed control by v/f method
L-30	Comparisons and analysis of different speed control methods
L-31	Numerical problems related to speed control of induction motor
L-32	Introduction to three phase synchronous machine and its construction details
L-33	Flux and mmfPhasors of three phase synchronous machine at different operating power factor motor
L-34	Phasor diagram of cylindrical rotor synchronous generator and motor
L-35	OCC and SCC characteristics of synchronous generator
L-36	OC and SC test on alternator
L-37	Zpfc and Potier triangle
L-38	Voltage regulation of alternator by emf method
L-39	Voltage regulation of alternator by mmf method
L-40	Numerical problems related to voltage regulation
L-41	Operating characteristics of alternator
L-42	Active and reactive Power output equations
L-43	Maximum power conditions
L-44	Power factor control of synchronous machine
L-45	V- curve of motor and generator
L-46	Two reaction theory of salient pole machine and phasor diagram
L-47	Power angle characteristics of synchronous machines
L-48	Hunting and damping in synchronous machine
L-49	**Single Phase Induction Motor construction and** Double revolving field theory,

L-50	Equivalent circuit, no load & block rotor tests on single phase induction motor
L-51	Starting methods of single phase induction motor

Name of the program	B.Tech.(ElectricalEngineering)	**Semester- IV**	**Year 2nd**
Name of Course:	Electronics		
Course Number:	EE 3405		
Core/Elective/other	**Core**		
Pre-requisites: Basic Knowledge of Electronics Devices and Circuits			
Course Outcomes			
4.	To understand basics Operational Amplifiers and Digital logic devices		
5.	To Analyse Simple and complex Circuits of Operational Amplifiers		
6.	Design Analog and digital circuits for electrical engineering applications		

Content	
1.	The differential and OP AMP basics, OP AMP applications- Instrumentation amplifier, current to voltage and voltage to current converter, OP AMP applications in Mathematical operations and analog computer. Schmitt trigger, Multivibrator circuits
2.	Feed back concept, Feedback topologies: voltage series current series, voltage shunt and current shunt, BarkhausenCriterion ,Wien bridge oscillator RC phase shift, Hartley, Clapp and Colpitt oscillator.
3.	Boolean algebra, Karnaugh Map upto 4 variables, logic gates, Logic Families
4.	Combinational Circuits: Half/ Full Adder, Half/ Full Subtractor, Adder-Subtractor, Decoder/Encoders, Multiplexer/Demultiplexer.
5.	Flip flops & their Application, Registers and Counters.
Text Books:	
5.	Millman&Halkies, Electronic Devices & Circuits, , McGraw Hill Education
6.	Ramakant A. Gayakwad, Op-Amp & Linear Integrated Circuits,
7.	William H.Gothmann, Digital Electronics, 9 edition, Pearson, July 28, 2011
8.	Thomas Floyd, Digital Fundamentals
9.	A. P. Malvino, Electronics Principles, McGraw-Hill Higher Education,
Reference Books :	
3.	Schaum's Outline Series, Electronics Devices & Circuits,
4.	Morris Mano, Digital Design, Pearson Education
URLs	
1.	https://nptel.ac.in/courses/108/102/108102112/
Lecture Plan	

Lecture No.	**Topic to be Covered**
L1.	Introduction to Diff Amp.
L2.	The differential and OP AMP basics,
L3.	OP AMP applications-1
L4.	OP AMP applications-2
L5.	OP AMP applications-3
L6.	Instrumentation amplifier,
L7.	Current to voltage and voltage to current converter,
L8.	OP AMP applications in analog computer.
L9.	Schmitt Trigger
L10.	Multi-vibrator circuits
L11.	Feed back concept,
L12.	Feedback topologies
L13.	Characteristics of Feedback topologies
L14.	Series Topologies
L15.	Shunt topologies

L16.	Sinusoidal Oscillators, Barkhausen Criterion
L17.	Wein bridge oscillator
L18.	RC phase shift oscillator.
L19.	Hartley and Colpitt Oscillator
L20.	Clapp Oscillator
L21.	Boolean algebra
L22.	Boolean Identiries
L23.	Karnaugh Map upto 4 variables-1
L24.	Karnaugh Map upto 4 variables-2
L25.	Logic Families
L26.	SOP and POS Implementation
L27.	Hardware design using Boolean function
L28.	AND-OR-INVERT Logic
L29.	Introduction to Combinational Circuits
L30.	Half Adder circuit
L31.	Full Adder Circuit
L32.	Half Subtractor
L33.	Full Suntractor circuits
L34.	Controlled Inverter
L35.	Adder-Subtractor
L36.	Decoder Circuit-1
L37.	Decoder Circuit-2
L38.	Encoder Circuit
L39.	Multiplexer/Demultiplexer
L40.	Introduction to Sequential Ciecuit
L41.	R-S Flip flop, D Type Flip flop
L42.	J-K Flip flop, Master-Slave Flip flop
L43.	Registers
L44.	Modulo N Counter -1
L45.	Modulo N Counter -2

B,Tech. (Electrical), 3rd Year

Fifth Semester

Name of Program	B.Tech (Electrical Engg.)	**Semester - V**	**Year –3rd**
Name of Course	Power Plant Engineering		
Course Code	ME3501		
Core / Elective / Other	**Core**		
Prerequisite if any:			